

# Piyush Datta

Website: <https://www.piyushdatta.com>

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## Work Experience

- Meta** San Francisco Bay Area, CA  
Software Engineer 2025 – Present
- Recruited from Ads Infrastructure to a high-priority AI team after delivering high-impact ML infrastructure work.
  - Designed large-scale ML calibration systems that increased revenue by ~\$10M/year and reduced fleet CPU usage by 2–3% across services handling millions of RPS.
  - Built infrastructure for a generalized ad-ranking model that improved ad value by 1–2% per ranked ad.
- Amazon Web Services** San Francisco Bay Area, CA  
Software Engineer 2020 – 2025
- Database kernel (MySQL/PostgreSQL) engineer on the AWS Aurora Database Engine team using C/C++.
  - Core engineer on the new *Aurora Limitless* product, presented at 2024 re:Invent (<https://aws.amazon.com/about-aws/whats-new/2023/11/amazon-aurora-limitless-database/>).
  - Led initiatives to create a sharded distributed storage engine capable of handling more than 1M TPS.
- Givex** Toronto, ON  
Software Engineer 2019 – 2020
- Implemented POS and transaction-based solutions using Python, JavaScript, C/C++, PostgreSQL, and Unix.

## Contributions & Projects

- 1<sup>st</sup> Place – Google Technology Prize, Hack for Social Impact 2025** <https://devpost.com/software/doorwai>  
*Python, React Native, Node/Express, Google Cloud*
- Created a fully functional cross-platform app that serves as an AI assistant for youth facing homelessness.
  - Enabled users to chat with the assistant via text or speech and place phone calls to agencies without exposing the user's phone number.
- 1<sup>st</sup> Place – Novice Champion, LLM Chess Hackathon (StrongCompute)** <https://github.com/PiyushDatta/chess-hackathon>  
*Python, PyTorch, TensorBoard*
- Trained a 3M-parameter vision-LLM agent from scratch using 8xH100s, FlashAttention, FSDP, GELU, and AMP.
  - Won Novice Champion after my LLM agent defeated other LLM agents in one-on-one chess matches with a 100-move limit.
- Amazon Patent (Pending) – AWS Aurora AI/RL-Based Database Configuration System**
- Lead designer and inventor of a database configuration system that uses reinforcement learning to predict optimal configurations based on customer workloads.
- Contributor to TensorFlow – Google Brain Open-Source Project** <https://www.tensorflow.org>  
*C/C++, Python*
- Contributed C++/Python changes to TensorFlow runtime/model-training infrastructure.

## Skills

**Technologies/Concepts:** C/C++, Rust, Python, JavaScript/TypeScript, Distributed Systems, IPC, 2PC, ML/AI, LLM, Reinforcement Learning, Consensus Algorithms, NewSQL/Distributed SQL

## Education

University of Toronto

Toronto, ON